

Lab 1 Solution

Objectives/Tested Concepts

1. Powering Devices + Connecting Interfaces (Cabling)
2. Assigning VLAN Access Ports
3. Configuring VLAN Trunk Ports
4. Switch Configuration
 - Hostname
 - Secret Password
 - Password Encryption
 - Console + VTY line password/login
 - No ip domain-lookup
5. Configuring Default Gateways
6. Using a L3 Switch for both L2 and L3 functionalities
 - Setting dot1q encapsulation for trunking
 - Using 'no switchport' to enable ip routing
 - Inter-VLAN Routing
7. Subnetting Networks with the /24, /25, /29, and /30 Masks
8. Determining IP Addressing/Subnets/Default Gateways
9. Configuring Port Security

Commands + (Some) Explanations

Powering Devices

Switches are just powered on all the time by default so the only thing to power are the end devices (PCs, servers, printer). To power a device, first click on the icon of the device and under the “Physical” tab (which is the tab you immediately see after clicking the icon) and find a rectangle + circle. A powered device has the rectangle filled in with a green color. If a device doesn’t have the green light, click on the circle right next to the rectangle (unless it’s a printer, then it’s the big blue button above).

Connections

Before doing any configurations, it’s best to do all of the cabling/connections (and powering devices) despite all of the common configurations listed before connections. Even though clicking the “Automatically Choose Connection Type” seems very tempting/easy, it might not connect the interface/port that is specifically listed, so you can either do all of your connections with “Copper Straight-Through” or “Copper Cross-Over” (this is all thanks to Auto MDI/MDIX – you don’t need to worry about this for the pkt).

FYI

For the MLS, I will be explaining pretty much every command and writing commands out fully. Everything after that will be shortened versions (what I do usually) and explanations will only be provided to VLAN relevant commands or not-so-basic-switch-configuration.

MLS

This is the Multi Layer Switch which means this device has the functionalities of both switches and routers (mostly). It's located in the middle of the pkt and connected to the "Internet". By default, all MLS (AKA L3 Switch) operate like a normal switch (AKA L2 Switch).

```
Switch>enable
```

```
Switch#configure terminal
```

```
Switch(config)#hostname MLS
```

```
MLS(config)#interface range GigabitEthernet 1/0/1-5
```

```
MLS(config-if-range)#switchport mode trunk
```

```
MLS(config-if-range)#switchport trunk allowed vlan 3,10,20,30,40
```

```
MLS(config-if-range)#exit
```

```
MLS(config)#interface GigabitEthernet1/0/24
```

```
MLS(config-if)#no switchport
```

```
MLS(config-if)#no shutdown
```

```
MLS(config-if)#ip address 100.52.10.253 255.255.255.252
```

```
MLS(config-if)#exit
```

```
MLS(config)#ip route 0.0.0.0 0.0.0.0 100.52.10.254
```

```
MLS(config)#interface vlan 3
```

```
MLS(config-if)#ip address 192.168.1.1 255.255.255.128
```

```
MLS(config-if)#interface vlan 10
```

```
MLS(config-if)#ip address 192.168.1.129 255.255.255.128
```

```
MLS(config-if)#interface vlan 20
```

```
MLS(config-if)#ip address 192.168.2.1 255.255.255.0
```

```
MLS(config-if)#interface vlan 30
```

```
MLS(config-if)#ip address 192.168.3.1 255.255.255.0

MLS(config-if)#interface vlan 40

MLS(config-if)#ip address 172.16.10.1 255.255.255.248

MLS(config-if)#vlan 3

MLS(config-vlan)#name Servers

MLS(config-vlan)#vlan 10

MLS(config-vlan)#name Office1

MLS(config-vlan)#vlan 20

MLS(config-vlan)#name Office2

MLS(config-vlan)#vlan 30

MLS(config-vlan)#name Office3

MLS(config-vlan)#vlan 40

MLS(config-vlan)#name Reception

MLS(config-if)#exit

MLS(config)#enable secret M0r3Po1nt5!

MLS(config)#line console 0

MLS(config-line)#password k33pLe8rn1ng

MLS(config-line)#login

MLS(config-line)#line vty 0 15

MLS(config-line)#password k33pLe8rn1ng

MLS(config-line)#login

MLS(config-line)#exit

MLS(config)#service password-encryption

MLS(config)#end

MLS#write
```

`enable` + `configure terminal` → to get into configuration mode

`hostname MLS` → set the hostname of the device to MLS

`interface range GigabitEthernet 1/0/1-5` → configure interfaces
G1/0/1, G1/0/2, G1/0/3, G1/0/4, and G1/0/5 at the same time

`switchport mode trunk` → set the interfaces to be a trunk

`switchport trunk allowed vlan 3,10,20,30,40` → allow traffic from
VLANs 3, 10, 20, 30, and 40 to go through the interface

`exit` → return to global configuration mode; optional

`interface GigabitEthernet 1/0/24` → configure interface G1/0/24

`no switchport` → make this link a L3 interface (for routing)

`no shutdown` → put link in the up/up state; enabling the interface

`ip address 100.52.10.253 255.255.255.252` → set the ip address to
the first usable ip address in 100.52.10.252/30

`exit` → return to global configuration mode

`ip route 0.0.0.0 0.0.0.0 100.52.10.254` → set a default route to
next hop 100.52.10.254

`interface vlan 3` → configure SVI (Switch Virtual Interface) for
VLAN 3 traffic

`ip address 192.168.1.1 255.255.255.128` → set the ip address to be
the first usable ip address in 192.168.1.0/25

`interface vlan 10` → configure SVI (Switch Virtual Interface) for
VLAN 10 traffic

`ip address 192.168.1.129 255.255.255.128` → set the ip address to be
the first usable ip address in 192.168.1.128/25

`interface vlan 20` → configure SVI (Switch Virtual Interface) for
VLAN 20 traffic

`ip address 192.168.2.1 255.255.255.0` → set the ip address to be the
first usable ip address in 192.168.2.0/24

`interface vlan 30` → configure SVI (Switch Virtual Interface) for
VLAN 30 traffic

`ip address 192.168.3.1 255.255.255.0` → set the ip address to be the
first usable ip address in 192.168.3.0/24

`interface vlan 40` → configure SVI (Switch Virtual Interface) for
VLAN 40 traffic

`ip address 172.16.10.1 255.255.255.248` → set the ip address to be
the first usable ip address in 172.16.10.0/29

`vlan 3` + `name Servers` → set name of vlan 3 to "Servers"

`vlan 10` + `name Office1` → set name of vlan 10 to "Office1"

`vlan 20` + `name Office2` → set name of vlan 20 to "Office2"

`vlan 30` + `name Office3` → set name of vlan 30 to "Office3"

`vlan 40` + `name Reception` → set name of vlan 40 to "Reception"

`exit` → return to global configuration mode

`enable secret M0r3P01nt5!` → set the secret password for Privileged
EXEC mode (from User EXEC mode; essentially to use `enable`
command)

`line console 0` → configure the console line

`password k33pLe8rn1ng` → set the console line password

`login` → require the password to be used

`line vty 0 15` → configure vty lines 0-15

`password k33pLe8rn1ng` → set the vty line password

`login` → require the password to be used

`exit` → return to global configuration mode

`service password-encryption` → encrypt passwords (if not already)

`end` → return to Privileged EXEC mode

`write` → another way of copying the running config to the startup
config

Office 1 Switch 1

This is the L2 switch that connects the servers to the rest of the network. Since the switch is connected to end devices, this is known as an “access switch” since it’s giving devices access to the network. It is switch on the very top left of the pkt.

```
Switch>en
```

```
Switch#conf t
```

```
Switch(config)#hostname Office1_Switch1
```

```
Office1_Switch1(config)#int range fa0/1-2
```

```
Office1_Switch1(config-if-range)#switchport mode access
```

```
Office1_Switch1(config-if-range)#switchport access vlan 3
```

```
Office1_Switch1(config-if-range)#int g0/1
```

```
Office1_Switch1(config-if)#switchport mode trunk
```

```
Office1_Switch1(config-if)#switchport trunk allowed vlan 3,10,20,30,40
```

```
Office1_Switch1(config-if)#vlan 3
```

```
Office1_Switch1(config-vlan)#name Servers
```

```
Office1_Switch1(config-vlan)#vlan 10
```

```
Office1_Switch1(config-vlan)#name Office1
```

```
Office1_Switch1(config-vlan)#vlan 20
```

```
Office1_Switch1(config-vlan)#name Office2
```

```
Office1_Switch1(config-vlan)#vlan 30
```

```
Office1_Switch1(config-vlan)#name Office3
```

```
Office1_Switch1(config-vlan)#vlan 40
```

```
Office1_Switch1(config-vlan)#name Reception
```

```
Office1_Switch1(config-vlan)#exit
Office1_Switch1(config)#ip default-gateway 192.168.1.1
Office1_Switch1(config)#no ip domain-lookup
Office1_Switch1(config)#enable secret M0r3Po1nt5!
Office1_Switch1(config)#line con 0
Office1_Switch1(config-line)#password k33pLe8rn1ng
Office1_Switch1(config-line)#login
Office1_Switch1(config-line)#line vty 0 15
Office1_Switch1(config-line)#password k33pLe8rn1ng
Office1_Switch1(config-line)#login
Office1_Switch1(config)#service password-encryption
Office1_Switch1(config)#end
Office1_Switch1#wr
```

`en` + `conf t` → shorthand versions of `enable` and `configure terminal`

`int range fa0/1-2` → configuring access ports together. Int = interface fa = FastEthernet

`switchport mode access` → set ports as access ports

`switchport access vlan 3` → assign ports to VLAN 3

`int g0/1` → configure port connected to MLS (trunk)

`switchport mode trunk` → set port as a trunk port

`switchport trunk allowed vlan 3,10,20,30,40` → allow traffic from VLANs 3, 10, 20, 30, and 40

`vlan 3` + `name Servers` → set name of vlan 3 to "Servers"

`vlan 10` + `name Office1` → set name of vlan 10 to "Office1"

`vlan 20` + `name Office2` → set name of vlan 20 to "Office2"

`vlan 30` + `name Office3` → set name of vlan 30 to "Office3"

`vlan 40` + `name Reception` → set name of vlan 40 to "Reception"

`ip default-gateway 192.168.1.1` → set the default gateway to 192.168.1.1 (VLAN 3 SVI IP Address)

`no ip domain-lookup` → disable domain-lookup; don't make it so that the switch tries to contact a host when you make a typo

`wr` → shorthand version for "write" which is short for "copy run start"

Office 1 Switch 2

This is the L2 switch that connects the Admin PC to the rest of the network. Like Office 1 Switch 1, this is also an access switch. It is below Office 1 Switch 1 in the pkt.

```
Switch>en
```

```
Switch#conf t
```

```
Switch(config)#hostname Office1_Switch2
```

```
Office1_Switch2(config)#int fa0/1
```

```
Office1_Switch2(config-if)#switchport mode access
```

```
Office1_Switch2(config-if)#switchport access vlan 10
```

```
Office1_Switch2(config-if)#int g0/1
```

```
Office1_Switch2(config-if)#switchport mode trunk
```

```
Office1_Switch2(config-if)#switchport trunk allowed vlan 3,10,20,30,40
```

```
Office1_Switch2(config-if)#vlan 3
```

```
Office1_Switch2(config-vlan)#name Servers
```

```
Office1_Switch2(config-vlan)#vlan 10
```

```
Office1_Switch2(config-vlan)#name Office1
```

```
Office1_Switch2(config-vlan)#vlan 20
```

```
Office1_Switch2(config-vlan)#name Office2
```

```
Office1_Switch2(config-vlan)#vlan 30
```

```
Office1_Switch2(config-vlan)#name Office3
```

```
Office1_Switch2(config-vlan)#vlan 40
```

```
Office1_Switch2(config-vlan)#name Reception
```

```
Office1_Switch2(config-vlan)#exit
```

```
Office1_Switch2(config)#ip default-gateway 192.168.1.129
```

```
Office1_Switch2(config)#no ip domain-lookup
```

```
Office1_Switch2(config)#enable secret M0r3Po1nt5!
```

```
Office1_Switch2(config)#line con 0
```

```
Office1_Switch2(config-line)#password k33pLe8rn1ng
```

```
Office1_Switch2(config-line)#login
```

```
Office1_Switch2(config-line)#line vty 0 15
```

```
Office1_Switch2(config-line)#password k33pLe8rn1ng
```

```
Office1_Switch2(config-line)#login
```

```
Office1_Switch2(config)#service password-encryption
```

```
Office1_Switch2(config)#end
```

```
Office1_Switch2#wr
```

`en` + `conf t` → shorthand versions of `enable` and `configure terminal`

`int fa0/1` → configuring the access port.

`switchport mode access` → set ports as access ports

`switchport access vlan 10` → assign ports to VLAN 10

`int g0/1` → configure port connected to MLS (trunk)

`switchport mode trunk` → set port as a trunk port

`switchport trunk allowed vlan 3,10,20,30,40` → allow traffic from VLANs 3, 10, 20, 30, and 40

`vlan 3` + `name Servers` → set name of vlan 3 to "Servers"

`vlan 10` + `name Office1` → set name of vlan 10 to "Office1"

`vlan 20` + `name Office2` → set name of vlan 20 to "Office2"

`vlan 30` + `name Office3` → set name of vlan 30 to "Office3"

`vlan 40` + `name Reception` → set name of vlan 40 to "Reception"

`ip default-gateway 192.168.1.129` → set the default gateway to 192.168.1.129 (VLAN 10 SVI IP Address)

`no ip domain-lookup` → disable domain-lookup; don't make it so that the switch tries to contact a host when you make a typo

`wr` → shorthand version for "write" which is short for "copy run start"

Office 2 Switch

This is the L2 switch that connects Office PC 1 and Office PC 2 to the network. This is also an access switch. It is located below Office 1 Switch 2 in the pkt.

```
Switch>en
```

```
Switch#conf t
```

```
Switch(config)#hostname Office2_Switch
```

```
Office2_Switch(config)#int range fa0/1-2
```

```
Office2_Switch(config-if-range)#switchport mode access
```

```
Office2_Switch(config-if-range)#switchport access vlan 20
```

```
Office2_Switch(config-if-range)#int g0/1
```

```
Office2_Switch(config-if)#switchport mode trunk
```

```
Office2_Switch(config-if)#switchport trunk allowed vlan 3,10,20,30,40
```

```
Office2_Switch(config-if)#vlan 3
```

```
Office2_Switch(config-vlan)#name Servers
```

```
Office2_Switch(config-vlan)#vlan 10
```

```
Office2_Switch(config-vlan)#name Office1
```

```
Office2_Switch(config-vlan)#vlan 20
```

```
Office2_Switch(config-vlan)#name Office2
```

```
Office2_Switch(config-vlan)#vlan 30
```

```
Office2_Switch(config-vlan)#name Office3
```

```
Office2_Switch(config-vlan)#vlan 40
```

```
Office2_Switch(config-vlan)#name Reception
```

```
Office2_Switch(config-vlan)#exit
```

```
Office2_Switch(config)#ip default-gateway 192.168.2.1
Office2_Switch(config)#no ip domain-lookup
Office2_Switch(config)#enable secret M0r3Po1nt5!
Office2_Switch(config)#line con 0
Office2_Switch(config-line)#password k33pLe8rn1ng
Office2_Switch(config-line)#login
Office2_Switch(config-line)#line vty 0 15
Office2_Switch(config-line)#password k33pLe8rn1ng
Office2_Switch(config-line)#login
Office2_Switch(config)#service password-encryption
Office2_Switch(config)#end
Office2_Switch#wr
```

`en` + `conf t` → shorthand versions of `enable` and `configure terminal`

`int range fa0/1-2` → configuring access ports together.

`switchport mode access` → set ports as access ports

`switchport access vlan 20` → assign ports to VLAN 20

`int g0/1` → configure port connected to MLS (trunk)

`switchport mode trunk` → set port as a trunk port

`switchport trunk allowed vlan 3,10,20,30,40` → allow traffic from VLANs 3, 10, 20, 30, and 40

`vlan 3` + `name Servers` → set name of vlan 3 to "Servers"

`vlan 10` + `name Office1` → set name of vlan 10 to "Office1"

`vlan 20` + `name Office2` → set name of vlan 20 to "Office2"

`vlan 30` + `name Office3` → set name of vlan 30 to "Office3"

`vlan 40` + `name Reception` → set name of vlan 40 to "Reception"

`ip default-gateway 192.168.2.1` → set the default gateway to 192.168.2.1 (VLAN 20 SVI IP Address)

`no ip domain-lookup` → disable domain-lookup; don't make it so that the switch tries to contact a host when you make a typo

`wr` → shorthand version for "write" which is short for "copy run start"

Office3_Switch

This is the L2 switch that connects Office PC 3 and the Printer to the network. This is also an access switch. It is located below the Office 2 Switch in the pkt.

```
Switch>en
```

```
Switch#conf t
```

```
Switch(config)#hostname Office3_Switch
```

```
Office3_Switch(config)#int range fa0/1-2
```

```
Office3_Switch(config-if-range)#switchport mode access
```

```
Office3_Switch(config-if-range)#switchport access vlan 30
```

```
Office3_Switch(config-if-range)#int fa0/2
```

```
Office3_Switch(config-if)#switchport port-security
```

```
Office3_Switch(config-if)#switchport port-security maximum 1
```

```
Office3_Switch(config-if#switchport port-security mac-address {Printer's MAC  
address}
```

```
Office3_Switch(config-if)#switchport port-security violation restrict
```

```
Office3_Switch(config-if)#int g0/1
```

```
Office3_Switch(config-if)#switchport mode trunk
```

```
Office3_Switch(config-if)#switchport trunk allowed vlan 3,10,20,30,40
```

```
Office3_Switch(config-if)#vlan 3
```

```
Office3_Switch(config-vlan)#name Servers
```

```
Office3_Switch(config-vlan)#vlan 10
```

```
Office3_Switch(config-vlan)#name Office1
```

```
Office3_Switch(config-vlan)#vlan 20
```

```
Office3_Switch(config-vlan)#name Office2
Office3_Switch(config-vlan)#vlan 30
Office3_Switch(config-vlan)#name Office3
Office3_Switch(config-vlan)#vlan 40
Office3_Switch(config-vlan)#name Reception
Office3_Switch(config-vlan)#exit
Office3_Switch(config)#ip default-gateway 192.168.3.1
Office3_Switch(config)#no ip domain-lookup
Office3_Switch(config)#enable secret M0r3Po1nt5!
Office3_Switch(config)#line con 0
Office3_Switch(config-line)#password k33pLe8rn1ng
Office3_Switch(config-line)#login
Office3_Switch(config-line)#line vty 0 15
Office3_Switch(config-line)#password k33pLe8rn1ng
Office3_Switch(config-line)#login
Office3_Switch(config)#service password-encryption
Office3_Switch(config)#end
Office3_Switch#wr
```

`en` + `conf t` → shorthand versions of `enable` and `configure terminal`

`int range fa0/1-2` → configuring access ports together. Int = interface fa = FastEthernet

`switchport mode access` → set ports as access ports

`switchport access vlan 30` → assign ports to VLAN 30

`int fa0/2` → configure port connected to Printer

`switchport port-security` → enable port security on this link

`switchport port-security maximum 1` → only allow 1 MAC address on this link

`switchport port-security mac-address {Printer's MAC address}`

→ manually configure the allowed MAC address on the link (click on the printer → config tab → FastEthernet0 to find MAC address)

`switchport port-security violation restrict` → set violation mode to restrict (drop traffic from unknown MAC address + record log)

`int g0/1` → configure port connected to MLS (trunk)

`switchport mode trunk` → set port as a trunk port

`switchport trunk allowed vlan 3,10,20,30,40` → allow traffic from VLANs 3, 10, 20, 30, and 40

`vlan 3` + `name Servers` → set name of vlan 3 to "Servers"

`vlan 10` + `name Office1` → set name of vlan 10 to "Office1"

`vlan 20` + `name Office2` → set name of vlan 20 to "Office2"

`vlan 30` + `name Office3` → set name of vlan 30 to "Office3"

`vlan 40` + `name Reception` → set name of vlan 40 to "Reception"

`ip default-gateway 192.168.3.1` → set the default gateway to
192.168.3.1 (VLAN 30 SVI IP Address)

`no ip domain-lookup` → disable domain-lookup; don't make it so that
the switch tries to contact a host when you make a typo

`wr` → shorthand version for "write" which is short for "copy run
start"

Reception Switch

This is the L2 switch that connects Reception PC 1 and Reception PC 2 to the network. This is also another access switch. It is located on the far right side of the pkt.

```
Switch>en
```

```
Switch#conf t
```

```
Switch(config)#hostname Reception_Switch
```

```
Reception_Switch(config)#int range fa0/1-2
```

```
Reception_Switch(config-if-range)#switchport mode access
```

```
Reception_Switch(config-if-range)#switchport access vlan 40
```

```
Reception_Switch(config-if-range)#int g0/1
```

```
Reception_Switch(config-if)#switchport mode trunk
```

```
Reception_Switch(config-if)#switchport trunk allowed vlan 3,10,20,30,40
```

```
Reception_Switch(config-if)#vlan 3
```

```
Reception_Switch(config-vlan)#name Servers
```

```
Reception_Switch(config-vlan)#vlan 10
```

```
Reception_Switch(config-vlan)#name Office1
```

```
Reception_Switch(config-vlan)#vlan 20
```

```
Reception_Switch(config-vlan)#name Office2
```

```
Reception_Switch(config-vlan)#vlan 30
```

```
Reception_Switch(config-vlan)#name Office3
```

```
Reception_Switch(config-vlan)#vlan 40
```

```
Reception_Switch(config-vlan)#name Reception
```

```
Reception_Switch(config-vlan)#exit
```

Reception_Switch(config)#ip default-gateway 172.16.10.1

Reception_Switch(config)#no ip domain-lookup

Reception_Switch(config)#enable secret M0r3Po1nt5!

Reception_Switch(config)#line con 0

Reception_Switch(config-line)#password k33pLe8rn1ng

Reception_Switch(config-line)#login

Reception_Switch(config-line)#line vty 0 15

Reception_Switch(config-line)#password k33pLe8rn1ng

Reception_Switch(config-line)#login

Reception_Switch(config)#service password-encryption

Reception_Switch(config)#end

Reception_Switch#wr

`en` + `conf t` → shorthand versions of `enable` and `configure terminal`

`int range fa0/1-2` → configuring access ports together.

`switchport mode access` → set ports as access ports

`switchport access vlan 40` → assign ports to VLAN 40

`int g0/1` → configure port connected to MLS (trunk)

`switchport mode trunk` → set port as a trunk port

`switchport trunk allowed vlan 3,10,20,30,40` → allow traffic from VLANs 3, 10, 20, 30, and 40

`vlan 3` + `name Servers` → set name of vlan 3 to "Servers"

`vlan 10` + `name Office1` → set name of vlan 10 to "Office1"

`vlan 20` + `name Office2` → set name of vlan 20 to "Office2"

`vlan 30` + `name Office3` → set name of vlan 30 to "Office3"

`vlan 40` + `name Reception` → set name of vlan 40 to "Reception"

`ip default-gateway 172.16.10.1` → set the default gateway to 172.16.10.1 (VLAN 40 SVI IP Address)

`no ip domain-lookup` → disable domain-lookup; don't make it so that the switch tries to contact a host when you make a typo

`wr` → shorthand version for "write" which is short for "copy run start"

End Device Values/Configurations

Data Server 1

- IP Address: 192.168.1.51
- Subnet: 255.255.255.128
- Default Gateway: 192.168.1.1

Data Server 2

- IP Address: 192.168.1.52
- Subnet: 255.255.255.128
- Default Gateway: 192.168.1.1

Admin PC

- IP Address: 192.168.1.191
- Subnet: 255.255.255.128
- Default Gateway: 192.168.1.129

Office PC 1

- IP Address: 192.168.2.4
- Subnet: 255.255.255.0
- Default Gateway: 192.168.2.1

Office PC 2

- IP Address: 192.168.2.5
- Subnet: 255.255.255.0
- Default Gateway: 192.168.2.1

Office PC 3

- IP Address: 192.168.3.2
- Subnet: 255.255.255.0
- Default Gateway: 192.168.3.1

Printer

- IP Address: 192.168.3.254
- Subnet: 255.255.255.0
- Default Gateway: 192.168.3.1

Reception PC 1

- IP Address: 172.16.10.5
- Subnet: 255.255.255.248
- Default Gateway: 172.16.10.1

Reception PC 2

- IP Address: 172.16.10.6
- Subnet: 255.255.255.248
- Default Gateway: 172.16.10.1

Potential Answers to Questions

Q: Why is the address for the printer end in .254 and not .255?

A: .255 is the broadcast address for the network the printer is in (192.168.3.0/24) which hosts can't use. This is also true for the first address which is 192.168.3.0 since that is the network address.

Q: Why did you make me go through this pain of determining values :(

A: To encourage more thinking about the network and IP addressing + subnets (they are very important fundamentals in networking)

Q: My ping requests aren't successful. What should I do?

A: Check for a) correct destination IP address in your ping command
b) correct default gateway values c) access VLAN / allowed VLANs on trunk interfaces d) you enabled routing on the MLS with `ip routing`
e) correct subnet masks f) correct IP address assignment for the SVI on the MLS g) correct static default gateway on the MLS if trying to reach 8.8.8.8

Q: I can't use certain commands listed above for some reason

A: This is likely due to either typos, being in the wrong configuration mode, or not having typed in a command beforehand (like no switchport on G1/0/24 for the MLS)

Q: Do I need to do anything with the "Internet" cloud?

A: No.

Q: I don't understand a lot of these commands or have never seen them

A: This was designed as a sort of assessment on VLANs and using a L3 Switch to allow Inter-VLAN routing which also somehow led me to also put in switch configurations, determining values, and a default gateway for a L3 device. Just as a note, this was not designed for beginners; it was designed for people at an intermediate level who have seen these commands before and just want some review or a bigger lab which combines multiple concepts instead of isolated ones. In any case, I highly suggest either looking back at the Netacad material or use the Internet to find some answers (I personally use a combination of Google + Jeremy's IT Lab for the CCNA on YT + sometimes AI).

Q: What commands are useful for debugging?

A: ``show running-configuration`` → overall view of configurations made

``show interfaces trunk`` → show trunking information

``show vlan brief`` → show which interfaces are assigned what VLAN

``show ip interface brief`` → show IP addresses of interfaces

``do {debugging command}`` → use debugging command in global config

``show ip routing`` → show routing table for L3 devices